Throughput Prediction in Beyond 5G Networks Using Machine Learning

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Abstract

This study investigates the use of machine learning to predict total UE1 throughput in a Beyond 5G wireless network. Four models, Linear Regression, Random Forest, XGBoost, and Long Short-Term Memory (LSTM), are developed and compared. Each model uses the previous three total UE1 throughput measurements (t−1, t−2, and t−3), along with Channel Quality Indicator (CQI) and jitter, to predict the next total UE1 throughput value. Model performance is evaluated using Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2). Among the four models, XGBoost achieved the highest test R^2 of 0.6570 and the lowest test RMSE of 7,621.46 Bytes/s, while Random Forest achieved the lowest test MAE of 4,956.63 Bytes/s. These results show that the tree-based models provided the strongest overall performance for total UE1 throughput prediction.

1. Introduction

1.1 Background

Wireless networks have evolved from first-generation (1G) systems to modern 5G and Beyond 5G (B5G) networks, supporting applications such as cloud computing, the Internet of Things (IoT), autonomous systems, and multimedia services. As network traffic becomes more dynamic, maintaining reliable throughput becomes increasingly difficult. Throughput measures the amount of data successfully transmitted over a network and is a key indicator of Quality of Service (QoS). Predicting throughput can help improve resource allocation and maintain network performance under changing conditions. Machine learning offers an effective approach by learning patterns directly from historical network measurements.

1.2 Problem Statement

This study addresses short-term prediction of total UE1 throughput from recent throughput history, jitter, and CQI. The main question is which of four machine learning approaches, Linear Regression, Random Forest, XGBoost, or LSTM, best predicts the next throughput value under changing network conditions.

1.3 Research Objectives

The primary objective of this study is to evaluate the ability of machine learning models to predict future wireless network throughput using historical network measurements. The specific objectives are:

- Develop a machine learning pipeline for throughput prediction using a publicly available Beyond 5G dataset.
- Predict future total UE1 throughput using three historical total throughput measurements together with jitter and Channel Quality Indicator (CQI).
- Compare the performance of Linear Regression, Random Forest, XGBoost, and Long Short-Term Memory (LSTM) models.
- Evaluate each model using Mean Absolute Error, Mean Squared Error , Root Mean Squared Error, and the coefficient of determination.

1.4 Contributions

This paper makes the following contributions:

- Evaluates four regression approaches on the same chronological Beyond 5G throughput prediction task.
- Examines whether recent throughput, jitter, and CQI are sufficient for short-term total UE1 throughput prediction.
- Uses SHAP to interpret the best-performing model and identify which inputs drive its predictions.

2. Methodology

2.1 Dataset

This project uses the publicly available dataset accompanying Service-Aware Real-Time Slicing for Virtualized Beyond 5G Networks [1]. The dataset was collected from a cloud-native Beyond 5G (B5G) network built using the OpenAirInterface (OAI) platform and deployed on Kubernetes. The testbed integrates an AI-enabled control framework with the FlexRAN controller to support real-time, service-aware radio access network (RAN) slicing and Quality of Experience (QoE) optimization.

The dataset contains 48,600 time-series observations collected from three User Equipments (UE1, UE2, and UE3) operating under varying network conditions. Each observation includes application-level throughput measurements, Channel Quality Indicator (CQI), jitter, and additional network traffic statistics. This study focuses on UE1. Total UE1 throughput is calculated by summing the throughput measurements from the UE1 web-rtc, sipp, and web-server applications for each observation. This total throughput value is used as the prediction target.

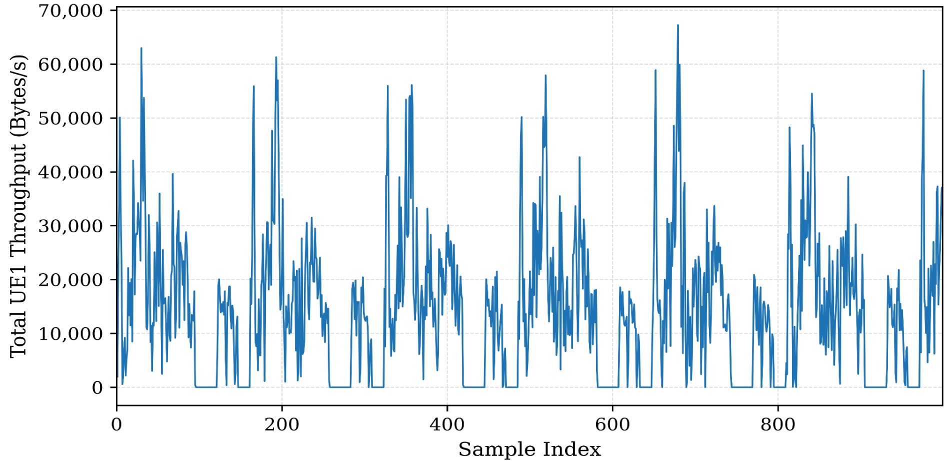

Figure 1. Total UE1 throughput over the first 1,000 samples. Total throughput was calculated by summing the WebRTC, SIPp, and web-server throughput values for UE1.

Figure 1 shows the variability of total UE1 throughput over the first 1,000 samples. Frequent throughput spikes and near-zero traffic periods highlight the difficulty of short-term throughput prediction.
Input Features:

- Total UE1 Throughput Lag 1
- Total UE1 Throughput Lag 2
- Total UE1 Throughput Lag 3
- UE1 Jitter
- UE1 Channel Quality Indicator (CQI)

Target Variable:

- Total UE1 Throughput

2.2 Data Preprocessing

After feature engineering, the dataset contained 48,597 usable observations. To preserve the temporal nature of the data, the samples were divided chronologically into 34,017 training samples (70%), 7,290 validation samples (15%), and 7,290 testing samples (15%). The training set was used to train each model, the validation set was used for hyperparameter tuning and model selection, and the testing set was reserved exclusively for the final performance evaluation.

Before training, all input features were normalized using Min-Max scaling. The scaler was fit only on the training set and then applied to the validation and test sets to prevent data leakage.

2.3 Feature Engineering

Three lag features were generated from total UE1 throughput at t−1, t−2, and t−3. The first three observations were removed because no complete lag history was available.

2.4 Machine Learning Models

2.4.1 Linear Regression

Linear Regression was used as the baseline model because it provides a simple and interpretable approach for predicting continuous values. Its performance serves as a benchmark for comparing the more advanced machine learning models. In addition, it was trained using the training set without hyperparameter tuning. The validation set was used for evaluation before testing the final model on the independent test set.

2.4.2 Random Forest

Random Forest constructs multiple decision trees and averages their predictions to model nonlinear relationships while reducing overfitting. Thirty hyperparameter combinations were evaluated by varying the number of trees (n_estimators), maximum tree depth (max_depth), and minimum leaf size (min_samples_leaf). The configuration with the lowest validation MAE was selected for the final model.

2.4.3 XGBoost

XGBoost is a gradient boosting algorithm that sequentially builds decision trees to reduce prediction error. Thirty hyperparameter combinations were evaluated on the validation set. The final model used 200 estimators, a learning rate of 0.05, a maximum depth of 6, a subsample ratio of 0.8, and a column subsample ratio of 1.0.

2.4.4 LSTM

The LSTM processes the three historical throughput measurements as a sequence, while jitter and CQI are provided as additional inputs. Thirty hyperparameter configurations were evaluated using the validation set. Early stopping selected the final model after 32 training epochs.

2.5 Evaluation Metrics

Four regression metrics were used to evaluate model performance: Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and the coefficient of determination (R^2). MAE measures the average absolute difference between predicted and actual throughput values. MSE calculates the average squared prediction error, placing a larger penalty on large errors. RMSE is the square root of MSE and expresses prediction error in the same units as throughput. R^2 measures the proportion of variation in total UE1 throughput explained by the model, with higher values indicating stronger predictive performance.

3. Results

3.1 Linear Regression Results

The Linear Regression model was first evaluated on the validation dataset and then on the independent test dataset. Table 1 summarizes its performance using MAE, MSE, RMSE, and R^2.

Table 1. Linear Regression Performance

Dataset	MAE (Bytes/s)	MSE (Bytes²/s²)	RMSE (Bytes/s)	R^2
Validation	6,060.58	72,216,468.23	8,498.03	0.5646
Test	6,098.99	73,535,470.54	8,575.28	0.5658

Validation and test performance remained similar. Linear Regression produced the largest errors among the four models, suggesting that a linear relationship was not sufficient to capture the patterns between the selected features and future throughput.

Figure 2 compares the actual and predicted total UE1 throughput values for the first 200 samples of the test dataset. The model captures some general changes in throughput but has difficulty following sudden fluctuations and large throughput spikes.

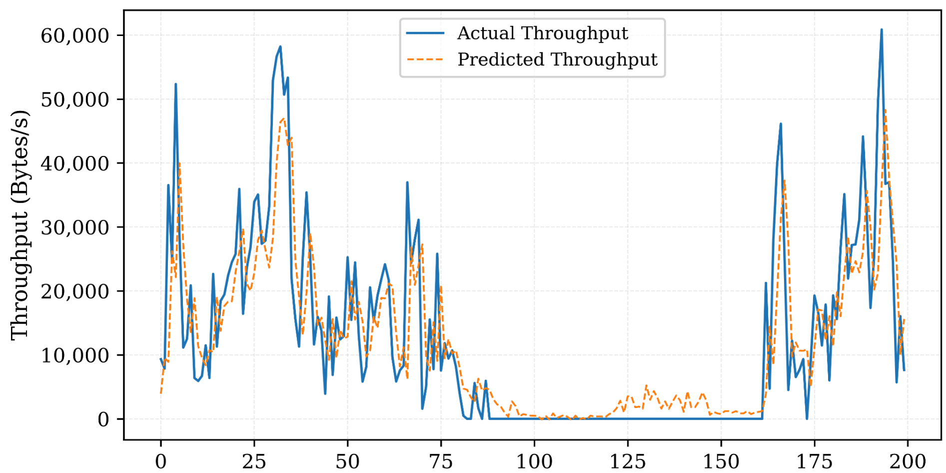

Figure 2. Actual and predicted total UE1 throughput for the first 200 samples of the test dataset using the Linear Regression model.

3.2 Random Forest Results

Hyperparameters were selected using the validation dataset. The final Random Forest model used 200 trees, a maximum depth of 10, and a minimum leaf size of 2. Table 2 summarizes the model's performance on the validation and test datasets.

Table 2. Random Forest Performance

Dataset	MAE (Bytes/s)	MSE (Bytes²/s²)	RMSE (Bytes/s)	R^2
Validation	4,941.68	57,930,333.75	7,611.20	0.6508
Test	4,956.63	58,135,049.10	7,624.63	0.6567

Random Forest showed little difference between validation and test performance. It substantially improved on the Linear Regression baseline and produced the lowest test MAE among the four models.

Figure 3 compares the predicted and actual total UE1 throughput values for the first 200 samples of the test dataset. Random Forest follows the overall throughput pattern more closely than Linear Regression but still has difficulty predicting some sudden throughput spikes and sharp changes between consecutive samples.

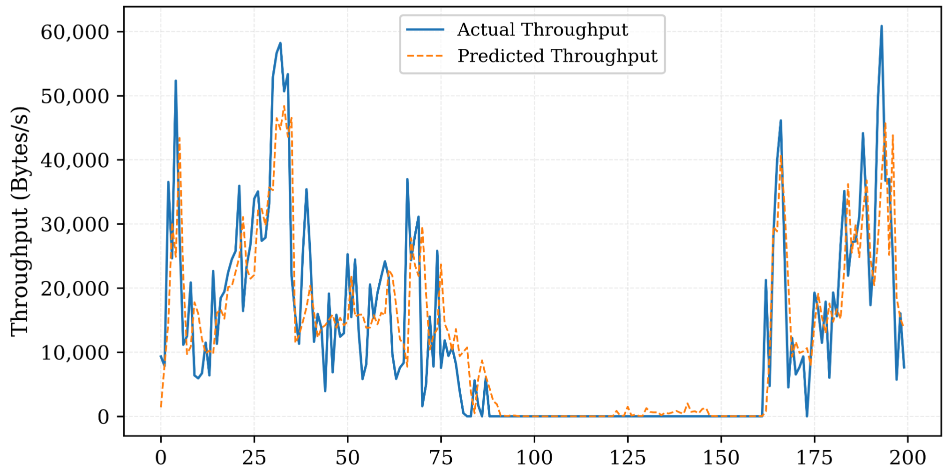

Figure 3. Actual and predicted total UE1 throughput for the first 200 samples of the test dataset using the Random Forest model.

3.3 XGBoost Results

Hyperparameters were selected using the validation dataset. The final XGBoost model used 200 estimators, a learning rate of 0.05, a maximum depth of 6, a subsample ratio of 0.8, and a column subsample ratio of 1.0. Table 3 summarizes the model's performance on the validation and test datasets.

Table 3. XGBoost Performance

Dataset	MAE (Bytes/s)	MSE (Bytes²/s²)	RMSE (Bytes/s)	R^2
Validation	4,986.64	57,873,927.00	7,607.49	0.6511
Test	4,987.59	58,086,630.55	7,621.46	0.6570

Validation and test performance were nearly identical. XGBoost produced the lowest test MSE and RMSE and the highest R^2 among the four models.

Figure 4 compares the actual and predicted total UE1 throughput for the first 200 samples of the test dataset. The model follows the overall throughput pattern across both active and near-zero traffic periods. However, it tends to underestimate some of the largest throughput spikes and produces errors when throughput changes sharply between consecutive samples.

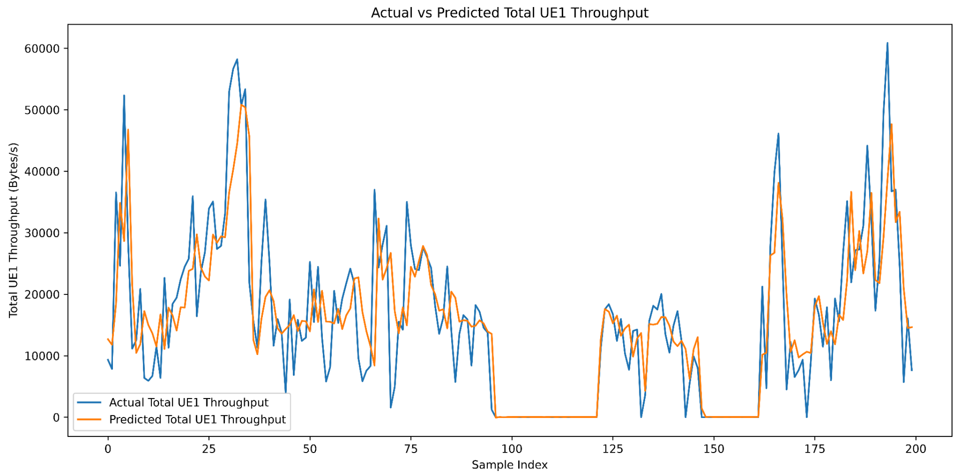

Figure 4. Actual and predicted total UE1 throughput for the first 200 samples of the test dataset using the XGBoost model.

3.4 LSTM Results

The LSTM model used the three historical throughput measurements as a sequential input, while jitter and CQI were provided as additional network features. The model was trained for up to 50 epochs, with early stopping used to prevent overfitting. Training stopped after 32 epochs. Table 4 summarizes the model's performance on the validation and test datasets.

Table 4. LSTM Performance

Dataset	MAE (Bytes/s)	MSE (Bytes²/s²)	RMSE (Bytes/s)	R^2
Validation	5,308.92	60,756,748.46	7,794.66	0.6337
Test	5,367.59	61,892,866.44	7,867.20	0.6345

The small difference between validation and test performance suggests limited overfitting. However, the LSTM remained less accurate than both tree-based models.

Figure 5 compares the actual and predicted total UE1 throughput for the first 200 samples of the test dataset. The LSTM follows the general throughput pattern and correctly identifies periods of near-zero traffic. However, its predictions are smoother than the actual measurements, causing it to underestimate several sharp throughput peaks. The model also produced slightly negative predictions during some periods of zero throughput.

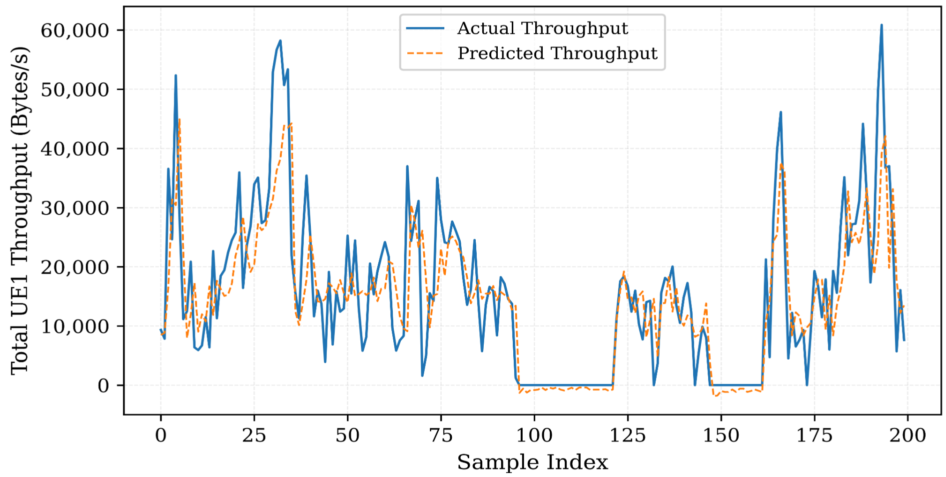

Figure 5. Actual and predicted total UE1 throughput for the first 200 samples of the test dataset using the LSTM model.

3.5 Comparative Analysis

Table 5. Comparison of Model Performance on the Test Dataset

Model	MAE (Bytes/s)	MSE (Bytes²/s²)	RMSE (Bytes/s)	R^2
Linear Regression	6,098.99	73,535,470.54	8,575.28	0.5658
Random Forest	4,956.63	58,135,049.10	7,624.63	0.6567
XGBoost	4,987.59	58,086,630.55	7,621.46	0.6570
LSTM	5,367.59	61,892,866.44	7,867.20	0.6345

XGBoost produced the lowest MSE and RMSE and the highest R^2, while Random Forest achieved the lowest MAE. Their performance was nearly identical and clearly exceeded the Linear Regression baseline, with LSTM falling between the linear and tree-based approaches. Because XGBoost had the strongest overall performance, it was selected for SHAP analysis to examine which features drove its predictions.

3.5.1 XGBoost Model Interpretation Using SHAP

Prediction accuracy alone does not explain why a model makes a particular prediction. Although XGBoost achieved the best overall performance, it consists of many decision trees that are difficult to interpret directly. SHAP (SHapley Additive exPlanations) was applied to quantify the contribution of each input feature and identify which network measurements had the greatest influence on the model's predictions [2].

Figure 6. Mean absolute SHAP values for the five input features used by the XGBoost model.

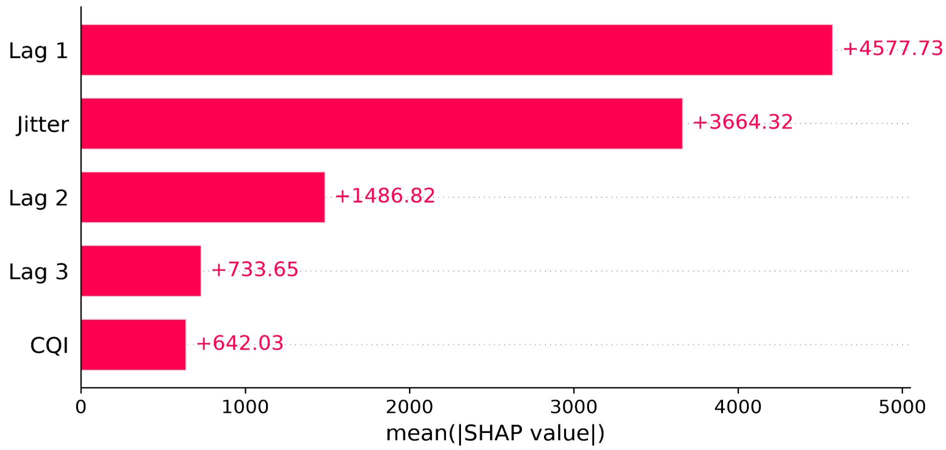

Figure 6 ranks the input features by their average contribution to the model's predictions. Lag 1 throughput was the most influential feature, followed by UE1 jitter, Lag 2 throughput, Lag 3 throughput, and CQI. The large difference between Lag 1 and the remaining features indicates that the most recent throughput measurement contains the strongest information for predicting the next throughput value. UE1 jitter was the second most influential feature, indicating that short-term variations in packet timing also contributed to future throughput predictions.

Figure 7. SHAP summary plot showing the contribution of each feature across all test samples. Point color represents the feature value, where blue indicates lower values and pink indicates higher values.

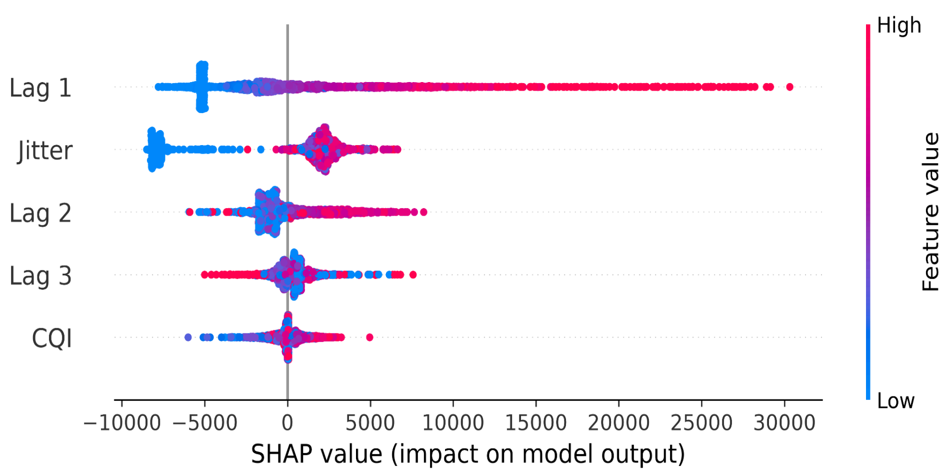

Figure 7 shows how each feature affected individual predictions across the test dataset. Each point represents one prediction, while the horizontal position indicates whether the feature increased or decreased the predicted throughput. High Lag 1 values generally produced positive SHAP values, increasing the prediction, whereas lower values reduced it. Jitter followed a similar pattern but with a smaller overall influence. Lag 2, Lag 3, and CQI contributed less consistently, reinforcing the feature ranking observed in Figure 6.

4. Discussion

The tree based models handled the dataset better than Linear Regression, suggesting that throughput depends on nonlinear relationships between the selected features. Random Forest and XGBoost also performed almost identically. This indicates that both methods were able to extract similar patterns from recent throughput, jitter, and CQI.

The LSTM did not gain the expected advantage from modeling throughput as a sequence. Its predictions often smoothed sharp changes instead of following them. One possible reason is the short input sequence: only three previous throughput measurements were available to the model. That may not provide enough temporal context to learn changes that develop over longer periods. Moreover, SHAP also showed that importance declined across the three lag features as the throughput measurements became older. This supports the use of recent throughput history for short-term prediction, but it also suggests that extending the historical window may have diminishing value. CQI contributed less than recent traffic behavior for this dataset.

The study is limited to UE1, five input features, and offline evaluation on one dataset. As a result, the findings cannot yet establish how the models would perform across different users, traffic conditions, or a live network.

5. Conclusion

This study compared four approaches for short-term total UE1 throughput prediction using recent throughput, jitter, and CQI. Random Forest and XGBoost produced the strongest results, with XGBoost achieving the best overall performance. SHAP analysis showed that its predictions depended most heavily on the most recent throughput measurement and UE1 jitter. The results indicate that a small set of recent network measurements can support useful throughput prediction, but sudden traffic changes remain difficult to capture. Testing this approach across more users and live network conditions is the next step toward determining its value for network resource management. Future work should test the models across multiple UEs and network conditions to determine whether the results generalize beyond UE1. Adding latency, packet loss, signal strength, or longer throughput histories may also improve predictions during sharp traffic changes. A final step would be real-time evaluation, where prediction latency and usefulness for resource allocation or congestion management can be measured directly. SHAP could then be applied across models and network conditions to determine whether the same features remain influential.

References

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